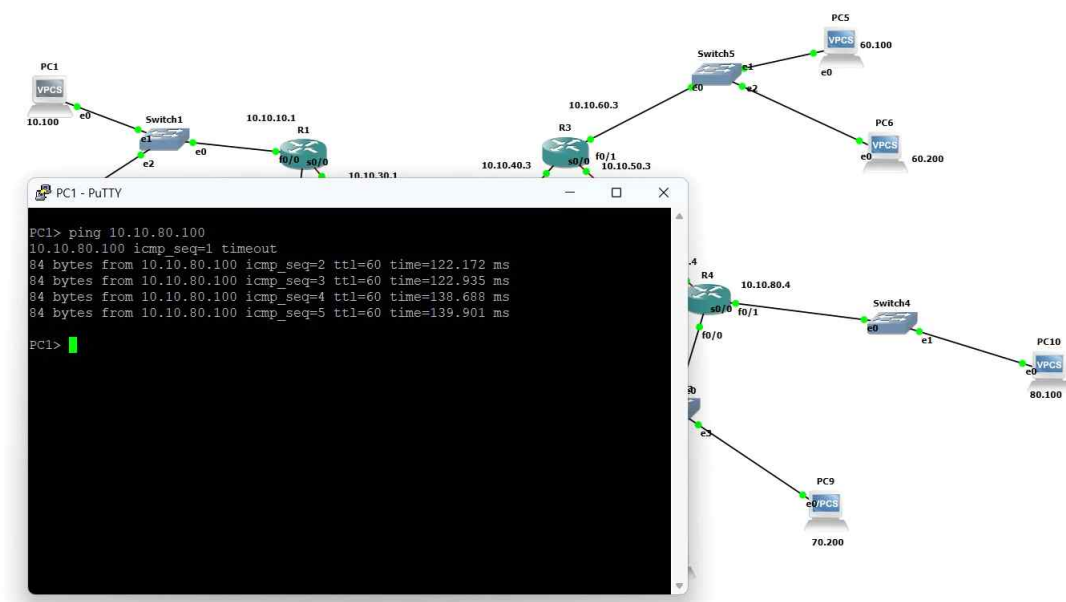


1. 모든 구간 통신가능 PC1 -> PC10통신



2. DACL 접근제어

```

R3#telnet 10.10.30.1
Trying 10.10.30.1 ... Open

User Access Verification

Username: testuser
Password:
[Connection to 10.10.30.1 closed by foreign host]
R3#ping 10.10.10.100

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.100, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 68/92/116 ms
R3#
  
```

3. ZFW 정책

```
R4(config)#do show zone security
zone self
  Description: System defined zone

zone inside
  Member Interfaces:
    FastEthernet0/0

zone outside
  Member Interfaces:
    Serial0/0

zone dmz
  Member Interfaces:
    FastEthernet0/1

R4(config)#do show zone-pair security
Zone-pair name in-out
  Source-Zone inside Destination-Zone outside
  service-policy policyall
Zone-pair name in-dmz
  Source-Zone inside Destination-Zone dmz
  service-policy policyall
Zone-pair name dmz-in
  Source-Zone dmz Destination-Zone inside
  service-policy policyicmp
Zone-pair name out-in
  Source-Zone outside Destination-Zone inside
  service-policy policyicmp

R4(config)#
```

4. wireshark 패킷 분석

The image shows a Wireshark packet capture window titled '*Standard input [R2 FastEthernet0/0 to R3 FastEthernet0/0]'. The packet list on the left shows a series of packets, with packet 153 selected. The packet details pane on the right shows the structure of the selected packet, which is an HTTP 400 Bad Request.

No.	Time	Source	Destination	Protocol	Length	Info
144	56.883307	10.10.40.3	10.10.40.2	TCP	60	23 → 48278 [FIN, PSH, ACK] Seq=241 Ack=75 Win=4054 Len=0
145	56.898844	10.10.40.2	10.10.40.3	TCP	60	48278 → 23 [ACK] Seq=75 Ack=242 Win=3888 Len=0
146	56.913873	10.10.40.2	10.10.40.3	TCP	60	48278 → 23 [FIN, PSH, ACK] Seq=75 Ack=242 Win=3888 Len=0
147	56.945389	10.10.40.3	10.10.40.2	TCP	60	23 → 48278 [ACK] Seq=242 Ack=76 Win=4054 Len=0
150	63.443398	10.10.40.2	10.10.40.3	TCP	60	64176 → 80 [SYN] Seq=0 Win=4128 Len=0 MSS=1460
151	63.458998	10.10.40.3	10.10.40.2	TCP	60	80 → 64176 [SYN, ACK] Seq=0 Ack=1 Win=4128 Len=0
152	63.473752	10.10.40.2	10.10.40.3	TCP	60	64176 → 80 [ACK] Seq=1 Ack=1 Win=4128 Len=0
153	63.488435	10.10.40.2	10.10.40.3	TCP	60	[TCP Dup ACK 152#1] 64176 → 80 [ACK] Seq=1 Ack=1 Win=0 Len=0
156	65.426542	10.10.40.2	10.10.40.3	HTTP	60	Continuation
157	65.653655	10.10.40.3	10.10.40.2	TCP	60	80 → 64176 [ACK] Seq=1 Ack=2 Win=4127 Len=0
165	80.127255	10.10.40.2	10.10.40.3	HTTP	60	Continuation
166	80.143052	10.10.40.3	10.10.40.2	TCP	176	80 → 64176 [ACK] Seq=1 Ack=4 Win=4125 Len=122 [T...
167	80.157858	10.10.40.3	10.10.40.2	HTTP	60	HTTP/1.1 400 Bad Request
168	80.172719	10.10.40.2	10.10.40.3	TCP	60	64176 → 80 [ACK] Seq=4 Ack=124 Win=4006 Len=0
169	80.188011	10.10.40.2	10.10.40.3	TCP	60	64176 → 80 [FIN, PSH, ACK] Seq=4 Ack=124 Win=4006 Len=0
170	80.203509	10.10.40.3	10.10.40.2	TCP	60	80 → 64176 [ACK] Seq=124 Ack=5 Win=4125 Len=0

Packet 153 details:

- [Time since reference or first frame: 1 minute, 5.42654200]
- Frame Number: 156
- Frame Length: 60 bytes (480 bits)
- Capture Length: 60 bytes (480 bits)
- [Frame is marked: False]
- [Frame is ignored: False]
- [Protocols in frame: eth:ethertype:ip:tcp:http:data]
- Character encoding: ASCII (0)
- [Coloring Rule Name: HTTP]
- [Coloring Rule String: http || tcp.port == 80 || http2]
- ▼ Ethernet II, Src: c0:02:46:3c:00:00 (c0:02:46:3c:00:00), Dst:
 - > Destination: c0:03:25:cc:00:00 (c0:03:25:cc:00:00)
 - > Source: c0:02:46:3c:00:00 (c0:02:46:3c:00:00)
 - Type: IPv4 (0x0800)
 - [Stream index: 6]
 - Padding: 0000000000
- ▼ Internet Protocol Version 4, Src: 10.10.40.2, Dst: 10.10.40.3
 - 0100 = Version: 4

5. rich-rule 정책 설정

```
gyeonggu@yeongguserver:~$ sudo firewall-cmd --permanent --add-rich-rule='rule family="ipv4" source address="10.10.70.200" protocol value="icmp" drop'
success
gyeonggu@yeongguserver:~$ sudo firewall-cmd --permanent --add-rich='rule family="ipv4" source address="10.10.70.150" service name="http" accept'
success
gyeonggu@yeongguserver:~$ sudo firewall-cmd --reload
success
gyeonggu@yeongguserver:~$ sudo firewall-cmd --list-rich-rules
sudo: firewall-cmd: command not found
gyeonggu@yeongguserver:~$ sudo firewall-cmd --list-rich-rules
rule family="ipv4" source address="10.10.70.200" protocol value="icmp" drop
rule family="ipv4" source address="10.10.70.150" service name="http" accept
gyeonggu@yeongguserver:~$
```